計算社會學
Computational Sociology

Spring 2024
National Taiwan University

授課老師：洪晨碩(研究室 414, 02-3366-1262, cshong@ntu.edu.tw)
上課時間：週三 14:20-15:20
約談時間：另約時間

Instructor: Chen-Shuo Hong
Class: Wednesday 7, 8, 9
Office hours: by appointment
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COURSE DESCRIPTION
Computational sociology (or computational social science) is an emerging interdisciplinary field that combines big data, computational methods, and social science. This upper-level course aims to introduce this field. Students will learn a variety of "big data" sources and computational methods and apply them ethnically to better understand human behavior and social phenomena. There are two modules. The first module will involve reading empirical work using various big data sources, including digital media/social media, administrative data, web-based/network experiments, social sensing, digital archives, and machine data. The second module will introduce computational methods, particularly machine learning and text analysis. We will also discuss some important extensions of contemporary artificial intelligence, such as generative AI. Some familiarity with mathematics and programming is encouraged but not required. Students will need to do empirical research independently or collaboratively as their final projects.

COURSE OBJECTIVES
At the conclusion of this course, you should:
1. Be familiar with data sources that are commonly used in computational sociology.
2. Understand how to apply computational methods to answer research questions regarding human behavior and social phenonmena.
3. Be able to assess computational sociology work critically.

REQUIREMENTS FOR THIS CLASS
Please read this syllabus carefully to familiarize yourself with the expectations of this class. There are some required textbooks for this class; all of them are freely available online. The remaining reading materials are research articles that will be posted to NTU COOL. Students are required to complete the readings before the class. While this course is about empirical investigations that apply computational sociology approach, students should be aware that some contents will be abstract and theoretical. This class will include a mix of student-lead discussion and and practice. Your class participation, group discussion, and final research projects will be important for your success.

It is understandable that students might need to miss a class for personal or family reasons. However, because class attendence is mandatory and you would need to take the responsibility to present your notes about the readings, you should request an exemption beforehand if needed.
Required textbooks:

1. Salganik, Matthew J. 2017. *Bit by Bit: Social Research in the Digital Age*. Princeton, NJ: Princeton University Press. Open review edition. https://www.bitbybitbook.com/.
2. Melanie Walsh, *Introduction to Cultural Analytics & Python*, Version 1 (2021), https://doi.org/10.5281/zenodo.4411250.

Recommended textbooks:
1. Wickham, Hadley, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd edition. (R4DS). O'Reilly Media, Inc. ISBN: 1491910399. https://r4ds.had.co.nz/.
2. Silge, Julia, and David Robinson. 2017. *Text Mining with R: A Tidy Approach*. O'Reilly Media. ISBN: 1491981652. https://www.tidytextmining.com/.

GRADING:
This course is designed to help students gain hands-on experience in applying computational sociology and the key to success to develop a "collaborative" learning space. As a result, class attendance are crucial, and you should actively participate in group activities. For final project, there are several milestones that will help you make small, but incredemental progress throughout the course.

Class participation and attendance: 40%
Final project (60%)
- Proposal (10%)
- Data collection and descriptive analysis (10%)
- Preliminary results (10%)
- Presentation (10%)
- Final paper (20%)

Letter grades will be used (A+: 90-100, A: 85-89, A-: 80-84, B+: 77-79; B: 73- 76; B-: 70-72; C+: 67-69: C: 63-66; C-: 60-62, F: <=59, X: 0).

INSTRUCTIONS FOR FINAL PROJECT:
Students should be expected to complete a project paper. This paper should be an empirical paper and based on your final presentation. Regarding writing a paper, there are two useful writing resources that are open to all students: NTU library (https://www.lib.ntu.edu.tw/) and Writing Education Center (https://www.awec.ntu.edu.tw/).

NTU ACADEMIC REGULATION:
Full Academic Regulations:
https://www.aca.ntu.edu.tw/WebUPD/acaEN/UAADRules/AcademicRegulations.pdf.
Students have the responsibility of conforming in all respects to that ethic, and should be aware that academic misconduct might result in severe penalties, including a zero on the assignment in question or the entire course. Further sanctions from the provost include censure, suspension, or expulsion. Academic dishonesty includes but is not limited to: cheating, fabrication, plagiarism, and facilitating dishonesty. Since students are expected to be familiar with the Academic Regulations and the commonly accepted standards of academic integrity, ignorance of such standards by itself is not sufficient evidence of lack of intent. Late homework assignment after the prescribed window will be deducted by 1 letter grade (e.g., According to NTU COOL timestamp, A+ become A if turning in one day late, A+ become A- if two days late, and so on).

STATEMENT ON AI

Following NTU AI policy, I will not prohibit using Generative AI. However, I insist that you should appropriately cite your AI tool in your work. If the generated contents include others' work, you should also use academic writing format (e.g., APA、MLA、Chicago Manual Style) to cite their work. It is your responsibility to check any possibilities of being considered academic dishonesty.

OTHER ACADEMIC SUPPORT

Your success in this course is important to me. Besides resources for writing and research, there are other useful resources.

The Center for Teaching and Learning Development provides a variety of resources and supports for students, including:
a) Academic Counseling Services : providing individual counseling and group tutoring for common courses and specialized courses by qualified peer tutors.
b) Online Resources: providing an online archive of past exams and problem-solving skills in Calculus, Statistics, and Physics.
For more information, please contact the Division of Learning Support, Center for Teaching and Learning Development. Tel: (02)33663367 Email: ntulsctld@ntu.edu.tw Location : 5th Floor, Liberal Education Classroom Building (N13 on the NTU map). Web: https://www.dlc.ntu.edu.tw/

The Center for Student Well-being and Student Counseling Center provides a multi-directional approach to the well-being of our students. It offers different kinds of services, including individual counseling, group counseling and workshops, mental health surveys, and psychological assessments. Please contact them through their websites (https://csw.ntu.edu.tw/Default.html).

ACCESSIBILITY RESOURCES

The school is committed to making reasonable, effective and appropriate accommodations to meet the needs of students with disabilities and help create a barrier free campus. If your disability requires an accommodation, please notify the instructors as early as possible in the course so that we may make arrangements in a timely manner.

COURSE SCHEDULE

Week (Date)	Topic	Reading	Group activities	Practicum
Week 1 (2/19)	Introduction	• Salganik, Matthew J. 2017. *Bit by Bit: Social Research in the Digital Age*. Princeton, NJ: Princeton University Press. Open review edition. Ch1 and Ch2. • Lundberg, Ian, Jennie E. Brand, and Nanum Jeon. 2022. "Researcher reasoning meets computational capacity: Machine learning for social science." *Social science research* 108: 102807.		Python environment
Module 1: Big data sources				
Week 2 (2/26)	Digital media/social media	• Zhong, W., Bailard, C., Broniatowski, D., & Tromble, R. (2024). Proud Boys on Telegram. *Journal of Quantitative Description: Digital Media*, 4, 1-47. https://doi.org/10.51685/jqd.2024.003 • Davidson, Thomas R., and Jenny Enos. 2024. "Engaging Populism? The Popularity of European Populist Political Parties on Facebook and Twitter, 2010–2020." *Political Communication* 42 (1): 171–200. doi:10.1080/10584609.2024.2369118.		Python basics
Week 3 (3/5)	Administrative data	• D. Tadmon, P.S. Bearman, Differential spatial-social accessibility to mental health care and suicide, *Proc. Natl. Acad. Sci.* U.S.A. 120 (19) e2301304120, https://doi.org/10.1073/pnas.2301304120 (2023). • Legewie, Joscha. 2016. Racial Profiling and Use of Force in Police Stops: How Local Events Trigger Periods of Increased Discrimination. *American Journal of Sociology* 122:2, 379-424 • (optional) 陳婉琪、王淑真、許哲維，2021，「不考試，公平嗎？以全國考招資料檢視多元入學公平性」，《台灣社會學刊》71：57-90。		Python basics / Data analysis
Week 4 (3/12)	Digital archives	• Becker, S. O., Hsiao, Y., Pfaff, S., & Rubin, J. (2020). Multiplex Network Ties and the Spatial Diffusion of Radical Innovations: Martin Luther's Leadership in the Early Reformation. *American Sociological Review*, 85(5), 857-894. https://doi.org/10.1177/0003122420948059 • Voyer, A., Kline, Z. D., Danton, M., & Volkova, T. (2022). From Strange to Normal:	• Proposal (10%)	Data analysis

		Computational Approaches to Examining Immigrant Incorporation Through Shifts in the Mainstream. *Sociological Methods & Research*, 51(4), 1540-1579. https://doi.org/10.1177/00491241221122596		
Week 5 (3/19)	Web-based experiment (guest speaker: 秦時雋)	• Schachter, A. (2016). From "Different" to "Similar": An Experimental Approach to Understanding Assimilation. *American Sociological Review*, 81(5), 981-1013. https://doi.org/10.1177/0003122416659248 • Salganik, Dodds, and Watts. "Experimental study of inequality and unpredictability in an artificial cultural market." *Science*.		Experimental design workshop
Week 6 (3/26)	Machine data	• Lockhart, J.W., King, M.M. & Munsch, C. 2023. Name-based demographic inference and the unequal distribution of misrecognition. *Nat Hum Behav* 7, 1084–1095. https://doi.org/10.1038/s41562-023-01587-9 • Schwemmer, C., Knight, C., Bello-Pardo, E. D., Oklobdzija, S., Schoonvelde, M., & Lockhart, J. W. (2020). Diagnosing Gender Bias in Image Recognition Systems. *Socius*, 6, 301-321. https://doi.org/10.1177/2378023120967171		Data collection
Week 7 (4/2)	Ethnics	• Salganik, Matthew J. 2017. *Bit by Bit: Social Research in the Digital Age*. Princeton, NJ: Princeton University Press. Open review edition. Ch 6.		Data collection
Module 2: Computational methods				
Week 8 (4/9)	Machine learning	• Sun, X. (2024). Supervised machine learning for exploratory analysis in family research. *Journal of Marriage and Family*, 86(5), 1468–1494. https://doi.org/10.1111/jomf.12973 • Friedman, S., & Reeves, A. (2020). From Aristocratic to Ordinary: Shifting Modes of Elite Distinction. *American Sociological Review*, 85(2), 323-350. https://doi.org/10.1177/0003122420912941	• Data collection and descriptive analysis / Research design (10%)	scikit-learn
Week 9 (4/16)	Machine learning and qualitative data	• Giebel, S., Alvero, A., Gebre-Medhin, B., & Antonio, A. L. (2022). Signaled or Suppressed? How Gender Informs Women's Undergraduate Applications in Biology and Engineering. *Socius*, 8. https://doi.org/10.1177/23780231221127537		scikit-learn

		• Li, Zhuofan, and Corey M. Abramson. "Ethnography and Machine Learning: Synergies and New Directions." *arXiv preprint* arXiv:2412.06087 (2024).		
Week 10 (4/23)	Network experiment (guest speaker: 江彥生)* *This topic is about data source	• Damon Centola ,The Spread of Behavior in an Online Social Network Experiment. *Science* 329,1194-1197(2010). DOI:10.1126/science.1185231 • Shirado, H., Christakis, N. Locally noisy autonomous agents improve global human coordination in network experiments. *Nature* 545, 370–374 (2017). https://doi.org/10.1038/nature22332		Experimental design workshop
Week 11 (4/30)	Text analysis – basics	• van Loon, Austin. "Three families of automated text analysis." *Social Science Research* 108 (2022): 102798. • Shi, Y., Kiley, K., & DiPietro, S. M. 2024. To the Extreme: Exploring the Rise of a Deviant Culture in a Misogynist Digital Community. *Socius*, 10. https://doi.org/10.1177/23780231241272681		Text analysis
Week 12 (5/7)	Text analysis – sentiment analysis	• Bail, C. A. (2012). The Fringe Effect: Civil Society Organizations and the Evolution of Media Discourse about Islam since the September 11th Attacks. *American Sociological Review*, 77(6), 855-879. https://doi.org/10.1177/0003122412465743 • Flores, René D. "Do Anti-Immigrant Laws Shape Public Sentiment? A Study of Arizona's SB 1070 Using Twitter Data." *American Journal of Sociology* 123, no. 2 (2017): 333–84. https://www.jstor.org/stable/26546008.	• Preliminary results / pilot study (10%)	Text analysis
Week 13 (5/14)	Text analysis – topic modeling	• Ian Kennedy, Chris Hess, Amandalynne Paullada, Sarah Chasins,, Racialized Discourse in Seattle Rental Ad Texts, *Social Forces*, Volume 99, Issue 4, June 2021, Pages 1432–1456, https://doi.org/10.1093/sf/soaa075 • Wetts, Rachel. "Money and meaning in the Climate Change Debate: Organizational Power, Cultural Resonance, and the shaping of American media discourse." *American*		Text analysis

		Journal of Sociology 129.2 (2023): 384-438.		
Week 14 (5/21)	Text analysis – word embedding	• Kozlowski, A. C., Taddy, M., & Evans, J. A. 2019. The Geometry of Culture: Analyzing the Meanings of Class through Word Embeddings. *American Sociological Review*, 84(5), 905-949. https://doi.org/10.1177/0003122419877135 • Kim, K., & Askin, N. (2024). Feature-Based Structures of Opportunity: Genre Innovation in the American Popular Music Industry, 1958 to 2016. *American Sociological Review*, 89(3), 542-583. https://doi.org/10.1177/00031224241246271		Text analysis
Week 15 (5/28)	Text analysis – generative AI	• Bail, Christopher A. (2024). "Can Generative AI improve social science?." *Proceedings of the National Academy of Sciences* 121.21: e2314021121. • Davidson, T. (2024). Start Generating: Harnessing Generative Artificial Intelligence for Sociological Research. *Socius*, 10. https://doi.org/10.1177/23780231241259651		Gen AI in python
Week 16 (6/4)	Final project presentation		• Final project presentation (10%)	
Final paper (20%) due 6/10				